Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Introducing Forces — ANSWERS

Kinematics describes *how* something moves. Forces explain *why*. Every topic in this unit comes back to one question: **what forces are acting on an object, and what is their combined effect?** This handout builds the toolkit you use to answer that question for the rest of Forces 2.

What is a force?

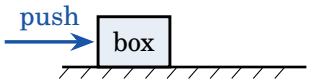
Key concept

A **force** is a **push** or a **pull** on an object, caused by it interacting with another object. Force is a **vector** — it has both a size and a direction — and it is measured in **newtons (N)**.

AQA Specification • §6.5.1.1 & 6.5.1.2 Forces as vectors • *P5 Forces*

aP5-4 A force is a push or a pull that acts on an object due to its interaction with another object.
aP5-7 Force is a vector quantity.

A force never exists on its own — it is always the result of *two* objects interacting. A hand pushes a box; the box is what feels the force. Every force this unit meets is one of two kinds, decided by a single question: **are the two objects touching?**



contact force — the hand touches the box

Contact and non-contact forces

AQA Specification • §6.5.1.2 Contact and non-contact forces • *P5 Forces*

aP5-5 Examples of contact forces include friction, air resistance, tension and normal contact force.
aP5-6 Examples of non-contact forces are gravitational force, electrostatic force and magnetic force.

Contact forces: the objects are physically touching. **Non-contact forces**: the objects are physically separated. Fill in as many examples of each as you can.

Contact forces	Non-contact forces
friction, air resistance, tension, normal contact force	gravitational force, electrostatic force, magnetic force

Worked example

A skydiver falling through the air feels **air resistance** (contact — the air touches them) and **weight** (non-contact — gravity acts without touching). Both are forces on the same object, from two different kinds of interaction.

Drawing a force diagram

AQA Specification • §6.5.1.4 Resultant forces • *P5 Forces*

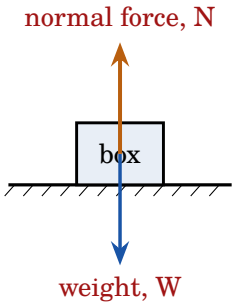
aP5-15 A number of forces acting on an object may be replaced by a single force with the same effect: the resultant force.
aP5-16 You should be able to calculate the resultant of two forces that act in a straight line.

A force diagram shows every force acting *on* one chosen object. Follow the method every time:

The method

1. Draw the object as a simple shape.
2. For **every** force acting *on* that object, draw an arrow from its centre.
3. The arrow's **direction** shows the force's direction; its **length** shows its relative size.
4. Label each arrow with the force's name.

A book resting on a table has two forces acting on it: its **weight**, pulling it down, and the **normal contact force** from the table, pushing it up. The book is not moving, so these two forces are **balanced** — equal in size, opposite in direction, and their resultant is zero.



A free-body diagram with forces that do not act along one line — and the HT skill of resolving them — is covered later in the course, not here.

Resultant forces

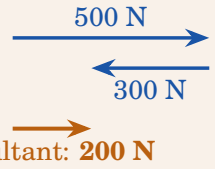
Along a straight line, combining forces is simple arithmetic: **add** forces pointing the same way, **subtract** forces pointing the opposite way.

Same direction



Two people push the same car the same way: $400 + 250 = 650$ N forward.

Opposite directions



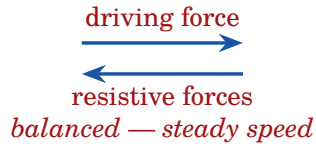
An engine drives forward at 500 N; resistance acts back at 300 N: $500 - 300 = 200$ N forward.

Newton's First Law

AQA Specification • §6.5.4.2.1 Newton's First Law • *P5 Forces*

aP5-63 If the resultant force on an object is zero: a stationary object stays stationary; a moving object continues at the same speed, in the same direction.
aP5-65 The velocity of an object only changes if a resultant force acts on it.

A steady speed does not mean *no* forces are acting — it means the forces are **balanced**. A car travelling at a constant speed has its driving force exactly cancelled by resistive forces (air resistance, friction). The resultant force is **zero**, so by Newton's First Law the car's velocity does not change.



balanced — steady speed

Newton's Second Law

AQA Specification • §6.5.4.2.2 Newton's Second Law • *P5 Forces*

aP5-68 The acceleration of an object is proportional to the resultant force acting on it, and inversely proportional to its mass: $F = ma$.

Newton's Second Law

$$\Sigma F = ma$$

force, F, in newtons (N) mass, m, in kilograms (kg) acceleration, a, in m/s^2

A bigger resultant force gives a **bigger acceleration**
The same force on a bigger mass gives a **smaller acceleration**

Worked example

A trolley of mass 4 kg is pushed with a resultant force of 20 N.

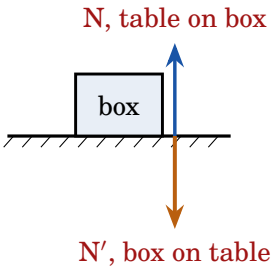
$a = F \div m = 20 \div 4 = \mathbf{5\ m/s^2}$

Newton's Third Law

AQA Specification • §6.5.4.2.3 Newton's Third Law • *P5 Forces*

aP5-71 Whenever two objects interact, the forces they exert on each other are equal in size and opposite in direction.

The test for a Third Law pair: it always acts on **two different objects**, at the same interface. The box's weight and the normal force on it (above) act on the *same* object, so they are **not** a Third Law pair — they are simply balanced. A genuine pair looks like this instead:

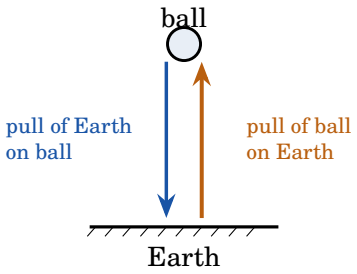


*Same contact point, equal size, opposite direction — one force on the **box**, one on the **table**.*

Putting it together: forces acting on an object

Newton's Laws only use the forces acting **on** the object you have chosen. A Third Law partner force acts on the *other* object, so it never belongs on the same force diagram.

Gravity is a Third Law pair too. The **Earth pulls the ball down**; the **ball pulls the Earth up**, with exactly the same size of force.



Why only the ball moves

The two forces are equal, but the Earth's mass is enormous next to the ball's. By $F = ma$, the same force gives the Earth an acceleration too small to ever notice — so only the ball appears to move.

Key terms

Glossary — write each definition in your own words

Force a push or a pull on an object, caused by it interacting with another object. Measured in newtons (N).

Vector a quantity with both a size and a direction. Force is a vector.

Contact force a force that needs the two objects to be touching — e.g. friction, tension, normal contact force.

Non-contact force a force that acts between objects that are separated — e.g. gravity, magnetism.

Resultant force the single force that has the same effect as all the forces acting on an object combined.

Newton's First Law zero resultant force $\Rightarrow$ no change in velocity.

Newton's Second Law $F = ma$ — resultant force equals mass times acceleration.

Newton's Third Law interacting objects exert equal and opposite forces on *each other*.